

Gavin Abrigo

(408)-888-8732 | abrigogavin05@gmail.com | [linkedin.com/in/gavinabrigo](https://www.linkedin.com/in/gavinabrigo) | gavinabrigo.com

SUMMARY

Student with experience in data and AI engineering, including Python-based ETL pipelines, RAG workflows and LLM-powered agents. Skilled in Python (Pandas, FastAPI), SQL and Postgres vector embeddings, with projects improving NLP model accuracy through knowledge graph retrofitting and automating image incident classification pipelines. Building scalable data infrastructure and AI agents.

EDUCATION

University of California Merced

Bachelor of Science in Computer Science & Engineering

May 2027

Merced, CA

TECHNICAL SKILLS

Programming Languages — TypeScript, JavaScript, GoLang, Java, Python, C/C++, HTML/CSS, LaTeX, MIPS Assembly, SQL, MATLAB, SPARQL

Data & Databases: — PostgreSQL, SQLite, MongoDB, Excel, Databricks

Frameworks & Tools: — Shell, React.js, FastAPI, Flask, REST APIs, Git, GitHub, Node.js, Next.js, Vite, StreamLit, Docker, CI/CD, JSON

AI & ML Tools — Nvidia NIM, Tesseract, OpenCV, PyTorch, Pandas, LLM API, Claude API, OpenAI API, Prompt Design

EXPERIENCE

LLM Knowledge Graph Retrofitter (KGR) - Prof. Pandey

Undergraduate Research Intern

Sep 2025 - Present

Merced, CA

- Improve factual accuracy to reduce hallucinations in NLP models. Building a Python-based Knowledge Graph Retrofitting (KGR) and Retrieval-Augmented Generation (RAG) pipeline using Wikidata and GraphRAG.
- Fine-tuning entity linking systems and evaluating LLM verification performance with NLP metrics such as exact match and F1 on benchmark datasets.

BART - Software Engineering Capstone

Software Engineering Associate

Jan 2026 - May 2026

Merced, CA

- Automatically classify infrastructure incident images submitted via web/mobile app, utilizing structured metadata, and route to appropriate maintenance teams
- Developing image embedding and similarity detection workflows (Google Gemini) to de-duplicate submissions with postgres vector embeddings, extract contextual location data, and reduce manual review overhead.

OculusIO (NFT Automation Software Startup)

Software Engineer Intern (backend)

Jan 2025 - Mar 2025

Remote

- Developed Bash and TypeScript CLI automation tools to streamline multi-language builds, integrating Go protobuf generation and client compilation—cutting setup time by 65% and build errors by 40%.
- Implemented cross-platform environment validation scripts for Windows, macOS, and Linux, standardizing workflows and improving developer onboarding efficiency across a 6-person team.

PROJECTS

SMART BART | Software Engineering Capstone | Python, Next.js, FASTAPI, AI fullstack | May 2026

- Image-based incident reporting system for Bay Area Rapid Transit (BART) that auto-classifies submitted maintenance photos and routes to correct department
- Gives maintenance workers a unified workflow for turning scattered photo reports into organized maintenance intelligence.
- Used vector embeddings, SentenceTransformers, and cosine similarity to cluster duplicate incident photos, so BART staff see one grouped report

[NYC Taxi Trip Duration Regression and Digit Classification](#) | Python, Scikit-learn, Pandas | Oct 2025 - Dec 2025

- Predicted total ride durations for NYC taxi trips, classified credit card fraud, and multi-digit MNIST images using Python, Pandas, and scikit-learn pipelines
- Building and evaluating classification and regression models across MNIST, Taxi Trip Duration,
- Implemented logistic regression with hyperparameter tuning and reusable data-preprocessing module; built XGBoost model with 5-fold stratified cross-validation, feature visualization, and outlier-cleaning using Python and Pandas